Curriculum Vitae

Léon Beaufils

Stanford, CA | USA
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Personal Website | GitHub: b-leon | LinkedIn

Research interests: Valuation of biodiversity and ecosystem services and design of market-based environmental instruments; environmental and climate economics; causal inference with spatial and remote sensing data

Education

Paris School of Economics — Paris, France

M.Sc. in Analysis and Policy in Economics (APE), Summa Cum Laude — *2023 – 2025*

- Average grade: 16.36/20 in Year 1, 16.79/20 in Year 2
- Thesis: *The Economic Cost of Pollinator Loss: Evidence from an Invasive Species*
  Advisor: *Matthew Gordon*, Grade: 18/20
- Selected courses: Microeconomics, Macroeconomics, Econometrics, Game Theory, Advanced Econometrics, Machine Learning in Economics, Energy Economics, Growth & Climate Change

University of Warwick — Coventry, UK

B.Sc. (Hons) in Economics, First Class Honours — *2020 – 2023*

- Thesis: *The Heterogeneous Impact of Right to Buy Policy on Residential Gas Consumption*
  Advisor: *Thiemo Fetzer*, Grade: 82%
- Selected courses: Microeconomics, Macroeconomics, Econometrics sequences, Mathematical Analysis, Mathematical Techniques B, Statistical Techniques B

Research Experience

ECHO Lab, Stanford Doerr School of Sustainability — Stanford, CA

Predoctoral Researcher (full-time), PI: Marshall Burke — *Oct. 2025 – present*

- Co-led construction and external validation of the **Global Harmonized Open Mortality Panel**, the largest existing harmonized panel of observed mortality and population data, spanning 115 countries and territories
- Assembled and harmonized mortality and population data from statistical agencies, published studies, and multilingual PDF and web sources; reconciled historical administrative boundaries and integrated census and survey data across administrative units and time periods
- Led much of the econometric analysis of the temperature–mortality relationship across 115 countries and territories, including regression estimation, results interpretation and visualization
- Responsible for the econometric analysis of two ongoing follow-up projects on how social safety net programs (India, Mexico, Brazil) and heating energy prices (US, Europe) moderate the temperature–mortality relationship, using distributed-lag non-linear panel fixed-effects models
- Managed and processed large-scale gridded climate, population and income datasets using R on a high-performance computing cluster
- Drafted manuscript sections and designed figures, tables and maps for two research papers

Department of Economics, University of Warwick — Coventry, UK

Research Assistant (part-time), PIs: Thiemo Fetzer & Ludovica Gazze — *Sep. 2022 – Sep. 2023*

- Constructed address-level datasets linking UK property-rights records, energy performance certificates, and energy consumption data using R; cleaned and analyzed the resulting data for research on residential energy efficiency

Energy & Environment Lab, University of Chicago — Chicago, IL

Research Assistant (full-time, remote), PI: Ludovica Gazze — *Jul. – Sep. 2022*

- Developed Stata code for stratified RCT randomization, including min-max re-randomization and power analysis, and designed information-provision treatment materials
- Scoped a prospective research project using California drought-risk and smart water meter data; contributed to final manuscript preparation of Browne et al. (2024), *Review of Economics and Statistics*

Research Papers

The Economic Cost of Pollinator Loss: Evidence from an Invasive Species *Feb. 2026*
Leon Beaufils
Working paper
Insect pollinators are in global decline, yet their economic value remains poorly quantified. This paper exploits the staggered spread of the Asian hornet (*Vespa velutina*) across France in an event-study design, using the heterogeneity-robust imputation estimator of Borusyak, Jaravel, and Spiess (2024) to identify causal effects on beekeeping, agricultural yields, and farmland values. I find that hornet invasion reduces beehive counts by 11.4%, oilseed rape yields by 5.2%, and farmland prices by 4.3%, suggesting sizable economic costs associated with pollinator loss.

Understanding and addressing temperature impacts on mortality *Oct. 2026*
Marshall Burke, Andrew Wilson, Ziqian Xia, ***Leon Beaufils****, Tumenkhusel Avirmed, Jonas Wallstein, Tina Bardot, Mariana Martins, and 12 co-authors*
Revise & Resubmit, Nature
Using nearly 500 million recorded deaths across 115 countries and territories, we estimate that heat and cold account for 3–15% of deaths across countries, with cold responsible for a larger share than heat in most settings. A systematic review of 52 causal studies evaluates evidence on interventions to reduce temperature-related mortality and identifies substantial gaps in their empirical support.

GHOMP: Global Harmonized Open Mortality Panel *Oct. 2026*
Andrew Wilson, ***Leon Beaufils****, Tumenkhusel Avirmed, Jonas Wallstein, Tina Bardot, and Marshall Burke*
Working paper
We construct a harmonized panel of observed mortality records and population data from as early as 1968, covering nearly 500 million recorded deaths across 16,833 subnational geography units in 115 countries and territories. The dataset provides daily, weekly, or monthly observations with temporally stable administrative boundaries. Upon release, GHOMP will be the largest openly available subnational mortality database with subannual temporal resolution.

Presentations

The Economic Cost of Pollinator Loss: Evidence from an Invasive Species
Graduate Social, Earth, and Environmental Data Science Seminar, Stanford *Mar. 2026*
E4C Summer School, Environmental Data Collection and Analysis for the Social Sciences, IP Paris *Jun. 2025*
Regulation, Environment & Markets Student Seminar, Paris School of Economics *May 2025*

Honors & Scholarships

Selected and funded participant, Energy4Climate (E4C), Institut Polytechnique de Paris (IP Paris) *2025*
Bourse sur critères sociaux, CROUS, need-based, highest tier *2023 – 2025*
Congratulatory letter, PSE APE directors, for Year 1; invitation to discuss post-master plans *2024*
Congratulatory letter, Director of Undergraduate Studies, Warwick Economics, for Year 2 *2022*
Bourse au mérite des lycéens, Auvergne-Rhône-Alpes Region, highest tier, for Baccalauréat performance *2019*

Service & Other Experience

Warwick Economics Summit Coventry, UK
Press & Communication Coordinator *2021 – 2023*

Department of Economics, University of Warwick Coventry, UK
Cohort Representative *2020 – 2021*

UNICEF School Project, University of Warwick Coventry, UK
Team Leader *2020 – 2021*

Schoolhouse Farm Vancouver, Canada
Farm worker, post-high-school gap year *2019 – 2020*

Skills & Languages

Programming: R (data.table, fixest, ggplot2, arrow, sf, terra, tidymodels), Stata, Python
Software & Computing: HPC (Stanford Sherlock), QGIS, Git, LaTeX, Quarto
Languages: French (native), English (fluent), German and Italian (basic)
GRE Score: Quantitative 170 (Maximum Grade), Verbal 166 (96th Percentile)